SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – SHORT ANSWER TEST (5 QUESTIONS)

Course Code:	ITA0509	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 1: Short Answer Test
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
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Section / Batch:	CSE	**Date:**	14/01/2026

Code of Conduct

I 192311008 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Aasvitha

QUESTIONS: 5 **Total Marks:50**

Annexure A

Q1. Smart Traffic Monitoring System

A smart city is planning to install cameras at major traffic junctions to monitor vehicles, pedestrians, and traffic conditions. The captured images may vary in resolution, lighting, and camera position.
Apply the fundamental concepts of Computer Vision to design a suitable processing pipeline for this traffic monitoring system. Explain the major stages from image acquisition to object detection and classification.

Q2. Automated Attendance System

A college wants to implement an automated attendance system using cameras that capture students sitting inside classrooms. The images may contain different lighting conditions, face orientations, shadows, and multiple students.
Analyze how image representation, preprocessing, feature extraction, and face recognition can be applied to develop the proposed attendance system. Discuss the major challenges that may affect its accuracy.

Q3. Medical Image Analysis System

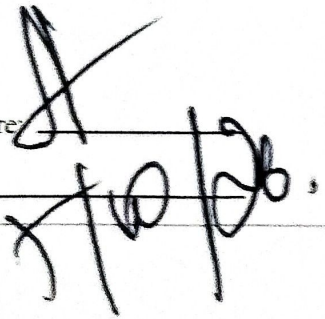
A hospital plans to use a Computer Vision system to assist doctors in analyzing X-ray images for identifying abnormal regions. The images may contain noise, poor contrast, and different brightness levels.
Evaluate the importance of image resolution, intensity, preprocessing, segmentation, and feature extraction in this application. Explain how these factors can influence the accuracy and reliability of the system.

Q4. Security Surveillance System

A shopping mall wants to develop an intelligent surveillance system using cameras installed at entrances, corridors, and parking areas. The system must identify people and objects under different lighting and crowded conditions.
Analyze how the fundamental stages of a Computer Vision system can be used for this application. Explain the effects of illumination, occlusion, camera position, and image quality on the final results.

Q5. Smart Agriculture Monitoring

A farming organization wants to use drone cameras to identify healthy plants, damaged plants, weeds, and disease-affected areas in large agricultural fields. The captured images may contain shadows, different lighting conditions, and variations in plant size.
Create a suitable Computer Vision pipeline for this application using image acquisition, preprocessing, segmentation, feature extraction, and classification. Justify the purpose of each stage.

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: 7/10/26	Date: _____	Date: _____

Traffic Monitoring System :-

A small Traffic Monitoring system uses computer vision techniques to identify vehicles, pedestrains and traffic conditions from camera images. It helps monitor traffic density, detect congestion and improve road safety.

* Processing pipeline :-

Image acquisition - Capture images or video using traffic cameras.

Image Representation - Represent images as pixel arrays with intensity or RGB values.

Preprocessing - Resize images, reduce noise, and improve contrast

Feature Extraction - Identify edges, shapes, textures and object patterns.

Object extraction - Locate vechicles and pedestrians using models such as YOLO.

Classification and analysis - Classify cars, buses, bikes and pedestrians; estimate traffic density.

* Challenges and solutions :-

↳ Different lighting conditions can be handled using Contrast enhancement.

↳ Different camera positions can be addressed through camera calibration.

Face detection :-

Detect individual faces using algorithms such as Haar cascades, HOG-based detectors or deep-learning face detectors.

↳ Feature extraction - Extract distinguishing facial features such as eye spacing, nose shape and facial patterns. Modern systems use deep-learning models to generate numerical face embeddings.

↳ Face recognition : Compare the extracted features ~~from~~ with registered student face templates stored in a database.

↳ Attendance recording : If the similarity score satisfies a suitable threshold, the system identifies the student and records attendance with the date and time.

≠ Challenges affecting accuracy :-

↳ Lighting variations : Poor lighting and shadows reduce facial visibility, image enhancement and appropriate lighting help address this.

↳ Face orientation : Side facing or tilted faces are difficult to recognize, face alignment and multiple angle training angles training images improve performance.

↳ Occlusion : Masks, hands, hair or other students may hide parts of a face.

↳ Multiple students : overlapping faces and crowded classr... make detection difficult.

↳ Image quality : Blurred or low - resolution images can reduce feature accuracy.

↳ Look - alike faces : Similar facial features can cause incorrect identification.

↳ Privacy and security : Student biometric data must be protected through access controls, encryption and appropriate consent.

3) Medical Image Analysis System :-

A medical image analysis system uses computer vision to analyse x-ray images and assist doctors in identifying abnormal regions such as fractures, infections, and other abnormalities. Image resolution, intensity, preprocessing, segmentation and feature extractions are important for accurate analysis.

→ Importance of different ranges :-

↳ Image resolution :- Resolution determines the amount of detail available in x-ray image. High - resolution images help identify small abnormalities, while low - resolution images may identify important details.

* Image intensity :- Identify represents the brightness of each pixel. In grayscale x-rays, different intensity levels distinguish bones, soft tissues and

uneven brightness. Techniques such as Gaussian filtering, histogram equalization and contrast enhancement improve image quality.

↳ Segmentation :- Segmentation divides an image into meaningful regions. It helps seperate bones, tissues and suspected abnormal areas. Thresholding and region-based methods can be used.

↳ Feature extraction :- Important features such as edges, shapes, textures, intensity variations are extracted from the image. These features help distinguish normal structures from possible abnormalities.

↳ classification :- Machine learning or deep learning such as convolutional Neural networks (CNNs), can classify images or regions as normal or potentially abnormal.

⇒ effect on accuracy and reliability :-

↳ Higher resolution improves the visibility of small abnormalities

↳ Improved contrast makes abnormal regions easier to distinguish

↳ Noise reduction prevents irrelevent variations from affecting analysis.

↳ Accurate segmentation isolates the region of interest.

↳ Relevent features improve classification performance.

→ Conclusion:-

Image resolution, intensity, preprocessing, segmentation and feature extraction work together to improve medical image analysis.

) Security Surveillance system:-

A ~~system~~ security Surveillance system uses computer vision to monitor ~~for~~ people and objects in shopping malls, entrances, corridors and parking areas. It can detect people, track movement, identify suspicious activities and improve security monitoring.

→ Fundamentals stages of the system:-

↳ Image acquisition: Cameras installed at the entrances, corridors and parking areas capture images and video continuously.

↳ Image representation: Captured images are represented as pixel matrices containing colour and intensity information.

→ Preprocessing: Noise removal, resizing, brightness correction, and constrast enhancement improve image quality.

Object detection: Computer vision models such as Yolo detect people, bags, vehicles and other relevant objects.

→ Feature extraction: The system extracts features such as shape, size, edges

as people, vehicles and personal belongings. classified into categories

Object tracking: Tracking algorithms follow detected objects across video frames to monitor movement and identify unusual activities

Alert generation: The system can notify security personal about predifined events, such as entry into restricted areas or unattended objects.

→ Effects of Environmental factors:-

Illumation → Use low-light cameras and contrast enhancement

Occulation → Use multiple cameras and object tracking

Camera position → Install cameras at suitable angles and calibrate them.

Image quality → Use high-resolution cameras and noise reduction

Crowded conditions → Use suitable detection and multiple-object tracking models.

5) Smart Agriculture Monitoring:-

A smart Agriculture Monitoring system uses drone cameras and computer vision to identify healthy plants, damaged plants, weeds and disease-affected regions. It helps farmers monitor large fields and take timely action.

→ Proposed Computer Vision Pipeline :-

↳ Image acquisition : Capture field image using drone cameras.

↳ Preprocessing :- Resize images, remove noise and correct brightness and contrast.

↳ Segmentation : Seperate plants, soil, weeds and damaged regions.

↳ Feature extraction : Extract colour, shape, texture and left-pattern features.

↳ classification :- classify regions as healthy plants, damaged plants, weeds or disease-affected plants.

↳ Result and Action :- Generate a field health map to guide irrigation, weed control, and crop treatment.

→ Challenges and solutions :-

↳ Shadows and varying sunlight can be handled through image normalization.

↳ Different plant sizes can be managed by training with images of different scales.

↳ Similar-looking weeds and crops require representative training data.

↳ Blurred images can be reduced by using suitable camera settings and image-quality checks.